

SMARTPHONE MARKET INTELLIGENCE ANALYSIS

Dataset: smartphones

Rows: 980

Columns: 26

LEVEL 1 – BASIC ANALYSIS

Q1. Total Smartphone Models

How many smartphone models exist in the dataset?

Q2. Brand Catalogue Overview

List all smartphone brands and the number of models offered by each brand. Sort by model count descending.

Q3. Budget 5G Phones

Find all smartphones priced below ₹15,000 that support 5G. Display brand, model, price, and rating.

Q4. Missing Rating Audit

How many smartphones have missing ratings? What percentage of the total catalogue does this represent?

Q5. Market Benchmark Metrics

Calculate the overall average:

- Price
- Rating
- Battery Capacity
- RAM
- Internal Storage

LEVEL 2 – BUSINESS ANALYSIS

Q6. Price Segment Distribution

Classify smartphones into:

- Budget (< ₹15,000)
- Mid-Range (₹15,000–₹40,000)
- Premium (₹40,000–₹80,000)
- Flagship (> ₹80,000)

Show:

- Number of phones
 - Average rating
- for each segment.**

Q7. Brand-wise Performance

For each brand, calculate:

- Average Price
- Average Rating

Include only brands with at least 5 smartphones.

Q8. Premium Brand Identification

Identify brands whose average smartphone price is above the overall market average.

Q9. Value-for-Money Brands

Find brands that provide above-average ratings while charging below-average prices.

Q10. Android vs iOS Comparison

Compare Android and iOS smartphones on:

- Average Price
- Average Rating
- Average Battery Capacity
- Average RAM

Q11. 5G Price Premium

Compare the average price of 5G phones versus non-5G phones.

Q12. 5G Adoption by Brand

For each brand, calculate:

- Total Models
- 5G Models
- 5G Adoption Percentage

Q13. Processor Market Share

Which processor brands power the most smartphones?

Q14. Fast Charging Brand Analysis

-- Brands with 3+ fast-charging models (non-NULL wattage).

Q15. Camera Comparison by Brand

-- Top 10 brands by average rear camera MP.

Q16. Refresh Rate vs Price Analysis

-- Custom buckets: 60 / 90 / 120 / 144Hz+

Q17. Expandable Storage Impact

-- Does extended memory support mean less built-in storage?

LEVEL 3 – INTERMEDIATE ANALYSIS

Q18. Hidden Gem Smartphones

-- Below market avg price AND above market avg rating.

Q19. Potentially Overpriced Smartphones

-- Above market avg price AND below market avg rating.

Q20. Battery Capacity Tiers and Ratings

-- Four tiers: Low / Standard / High / Ultra

Q21. NULL Handling in Fast Charging — Data Quality Check

-- Phones that claim fast charging available but wattage is NULL.

Q22. Processor Speed vs Price Bands

-- Avg speed, price, rating per processor brand. No NULLs.

Q23. RAM Tier Distribution

-- Group by RAM capacity; bucket anything outside common tiers.

Q24. Fully-Loaded Phones — Multi-Feature Filter

-- 5G + NFC + IR Blaster + Fast Charging + 120Hz+ + 8GB+ RAM

-- Also shows % of total catalogue.

Q25. High-Resolution Camera Smartphones

Which smartphones offer rear cameras exceeding 100 MP?

LEVEL 4 – ADVANCED SQL ANALYSIS

Q26. Highest-Rated Smartphone Per Brand

-- Uses ROW_NUMBER() partitioned by brand.

-- If two phones tie for top rating, both appear (use RANK).

Q27. Price Ranking Within Each Brand

-- RANK() with PARTITION BY brand; ties get the same rank.

Q28. Price Percentile Within Brand — Top Tier Flag

-- PERCENT_RANK() = 0 for cheapest, 1 for most expensive per brand.

-- Top 10% (PERCENT_RANK >= 0.90) flagged as 'Top Tier'.

Q29. Phones Priced Above Their Own Brand's Average

-- Classic correlated subquery pattern.

-- Inner query references outer query's brand_name.

Q30. Top 3 Rated Phones Per Brand — Handling Ties

-- DENSE_RANK() does not skip numbers after a tie.

-- e.g. 89, 89, 86 → ranks 1, 1, 2 (not 1, 1, 3)

Q31. LAG and LEAD — Price Gap Between Adjacent Phones

-- All 980 phones ordered by price ascending.

-- Shows each phone's price, the one below, the one above,

-- and the gap to the next phone up.

Q32. Brand Consistency Score — Coefficient of Variation

-- CV = STDDEV_POP(price) / AVG(price) * 100

-- Low CV = brand prices all models similarly (consistent)

-- High CV = brand spans many segments (diverse portfolio)

-- Only brands with 5+ phones included

Q33. Brand Ranking Within Price Segments (CTE + Window)

-- CTE: segment each phone and compute avg rating per brand-segment

-- Main SELECT: RANK() brands within each segment

Q34. Composite Smartphone Score — Leaderboard Top 20

- Formula (max values from dataset):
- $\text{rating} / 89 \times 40$ (weight: 40%)
- $\text{battery_capacity} / 22000 \times 20$ (weight: 20%)
- $\text{ram_capacity} / 18 \times 20$ (weight: 20%)
- $\text{primary_camera_rear} / 200 \times 20$ (weight: 20%)
- Excludes any phone with NULL in these four fields.

Q35. Multi-CTE Tier Dominance Analysis

- Which brand has the MOST models in each price segment?
- CTE 1 → segment every phone
- CTE 2 → count models per brand per segment
- CTE 3 → rank brands within each segment
- Final → show only the dominant brand (rank = 1) per segment